

Reading the relation icons

Relational N-Back · relation stream reference. Every icon in the app follows the single rule below, and so does assets/relations/relation-guide.png.

THE ONE RULE THAT DECODES EVERY ICON

An icon always shows **two different shapes**. Count their sides.

Fewer sides = first term · **More sides = second term**

Then read the picture as a sentence: **first term** → **relation** → **second term**.

Left versus right is randomised on every single draw and never means anything. Size, nesting and height carry the relation, and which shape gets the big role, the outer role or the higher role is decided **only** by side count.

SIDE VALUES

 Circle 1 side	 Triangle 3 sides	 Square 4 sides	 Pentagon 5 sides	 Hexagon 6 sides
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The same side values drive the Shape stream and the relation icons, so the table only has to be learned once. A circle counts as one side.

HOW TO DECODE AN ICON IN FOUR STEPS

1. **Count sides on both shapes.** The two shapes are never the same, so one always has fewer sides than the other.
2. **Name the terms.** Fewer sides = first term. More sides = second term.
3. **Read the arrangement,** which gives you the family: two shapes at the same size = equivalence, one drawn inside the other = containment, one clearly big and one clearly small = magnitude, diagonal overlap with one higher = sequence.
4. **Ask what the first term is doing.** Is it the outer or the inner shape? The big or the small one? The higher or the lower one? That answer picks the relation out of its family.

WORKED EXAMPLE

An icon shows a large circle with a small hexagon drawn inside it. Circle has 1 side, hexagon has 6, so the **circle is the first term**. The shapes are nested, so the family is **containment**. The first term is the outer shape, which reads **circle contains hexagon** — the relation is **Contains**. Swap the roles so the hexagon is outside and the circle inside, and the same pair reads **circle is inside hexagon**, which is **Inside**.

ALL EIGHT RELATIONS



Same **EQUIVALENCE · INVERSE: SAME**

Both shapes share a centre and are the same size, so neither one is a container and neither one is bigger.

The icon shown: Circle (1 side) versus Square (4 sides), so the circle is the first term and both are drawn at the same size on one centre — which reads **Circle is the same as square.**



Opposite **EQUIVALENCE · INVERSE: OPPOSITE**

Two equal-sized shapes held apart. Equal size marks the equivalence family; the gap separates Opposite from Same.

The icon shown: Square (4 sides) versus Pentagon (5 sides), so the square is the first term and both are drawn at the same size with a gap between them — which reads **Square is the opposite of pentagon.**



Contains **CONTAINMENT · INVERSE: INSIDE**

The FEWER-sided shape is the big outer one; the MORE-sided shape sits inside it.

The icon shown: Circle (1 side) versus Hexagon (6 sides), so the circle is the first term and the circle is the outer shape — which reads **Circle contains hexagon.**



Inside **CONTAINMENT · INVERSE: CONTAINS**

The FEWER-sided shape is the small inner one; the MORE-sided shape is the big outer one.

The icon shown: Circle (1 side) versus Pentagon (5 sides), so the circle is the first term and the circle is the inner shape — which reads **Circle is inside pentagon.**



Leads to **SEQUENCE · INVERSE: FOLLOWS**

The FEWER-sided shape is the HIGHER one. Only up and down matter; the diagonal leans either way at random.

The icon shown: Triangle (3 sides) versus Square (4 sides), so the triangle is the first term and the triangle is drawn higher — which reads **Triangle leads to square.**



Follows SEQUENCE · INVERSE: LEADS TO

The FEWER-sided shape is the LOWER one. Only up and down matter; the diagonal leans either way at random.

The icon shown: Pentagon (5 sides) versus Hexagon (6 sides), so the pentagon is the first term and the pentagon is drawn lower — which reads **Pentagon follows hexagon.**



Greater MAGNITUDE · INVERSE: LESS

The FEWER-sided shape is the BIG one. The side of the icon it sits on is random.

The icon shown: Circle (1 side) versus Hexagon (6 sides), so the circle is the first term and the circle is drawn bigger — which reads **Circle is greater than hexagon.**



Less MAGNITUDE · INVERSE: GREATER

The FEWER-sided shape is the SMALL one. The side of the icon it sits on is random.

The icon shown: Circle (1 side) versus Square (4 sides), so the circle is the first term and the circle is drawn smaller — which reads **Circle is less than square.**

FAMILIES AND INVERSES

FAMILY	MEMBERS	WHAT THE ICON LOOKS LIKE
Equivalence	Same, Opposite	Two shapes at the same size . Sharing one centre = Same; held apart = Opposite.
Containment	Contains, Inside	One shape drawn inside the other.
Sequence	Leads to, Follows	Two equal shapes overlapping diagonally , one higher than the other.
Magnitude	Greater, Less	Two shapes side by side, one clearly big and one clearly small .

WHAT THE RELATION PROMPT ASKS

PROMPT	PRESS THE RELATION BUTTON WHEN...
Same relation	the relation N trials back is the exact same relation as the current one.
Inverse	the relation N trials back is the inverse partner of the current one: Contains ↔ Inside, Leads to ↔ Follows, Greater ↔ Less.
Same family	both relations come from the same row of the family table above, whichever members they are.

Same and Opposite are their own inverses. Under an **Inverse** prompt they therefore behave exactly like a **Same relation** prompt, and they always satisfy **Same family** against each other.

TWO THINGS THAT ARE EASY TO CONFUSE

The white outline is not part of the relation

When the Shape stream is on, a white outline is drawn around the relation icon. That outline is the **Shape** stimulus and it has its own response button. Only the yellow shapes inside form the relation. An outline never counts as the container in Contains or Inside. Turning the Shape stream off removes the outline entirely and draws the relation icon at full size.

Big versus small is a role, not a position

Greater and Less both draw one big shape and one small shape side by side, and either one can appear on the left. The relation is decided purely by **which shape has more sides**: if the fewer-sided shape is the big one the relation is Greater, and if it is the small one the relation is Less. The same logic applies to outer versus inner in Contains and Inside, and to higher versus lower in Leads to and Follows.

Relational N-Back · relation icon reference. The rule lives in **SUBJECT_RULE** in `src/constants.js` and the first/second assignment in `src/symbols.js`; the icons in this document were generated from those modules.